Scientific Publication in SE, Seminar 5:

Evidence and Trust in our AI Future

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Truth Fix: (Pre-)Registered Reports

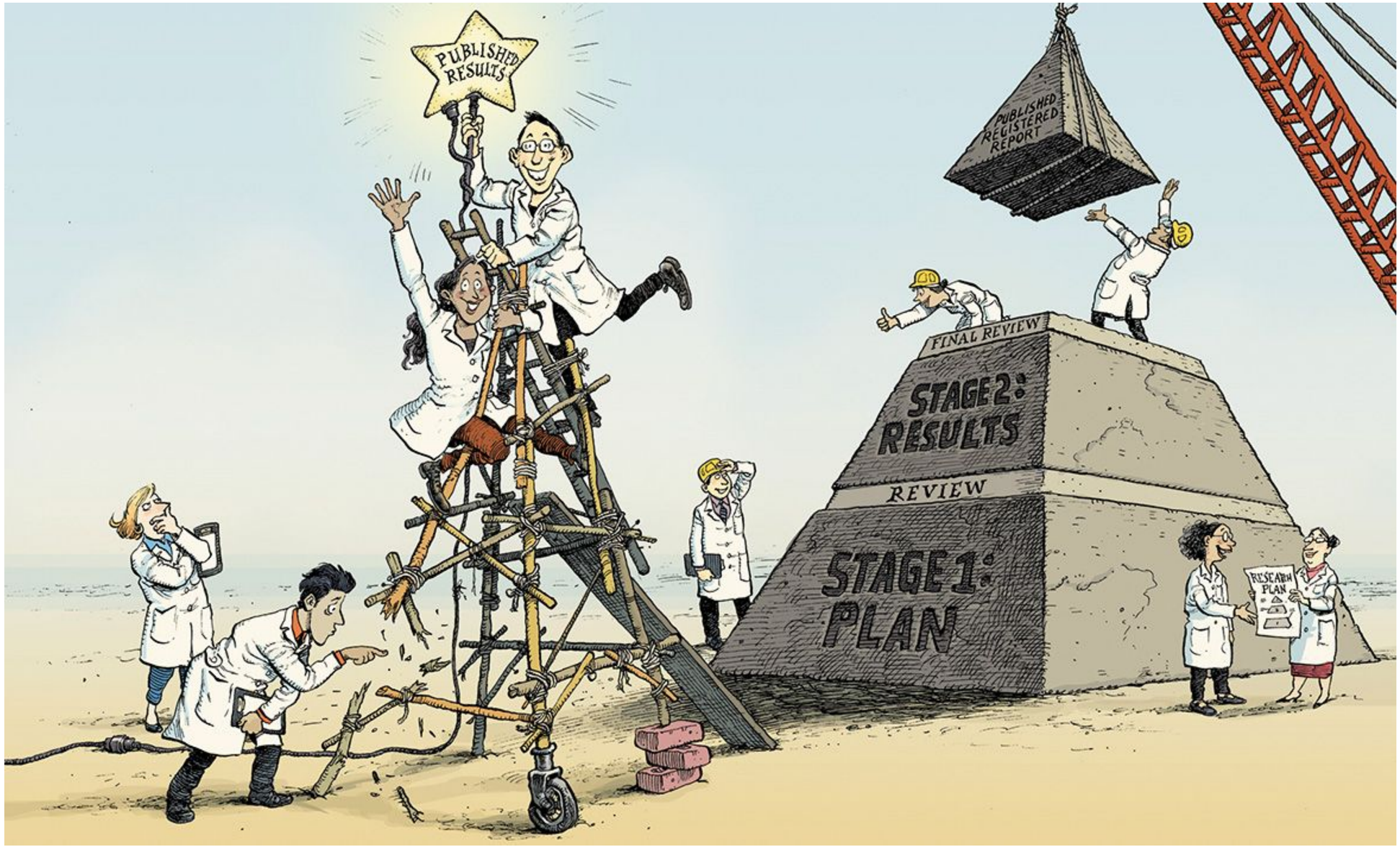


Illustration by David Parkins in Nature, September 2019

Truth Fix: (Pre-)Registered Reports



A form of self-blinding, next step after double blind!

200+ Journals today offer pre-registration!

Acceptance rate in stage 2: 90% (Cortex journal)

Null results: 66% RR replicat., 50% RR novel, 5-20% non-RR

Empirical Software Engineering (2023) 28:55
https://doi.org/10.1007/s10664-022-10277-5

EDITORIAL

Registered reports in software engineering

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Fabricated data!?

> Reviewing and claim-evidence checks tests if things **fit**, **not** whether the evidence is real

- Improved claim-evidence checking and AI-supported review can:
  - Fit and alignment: do the claims follow from the evidence?
  - Catch: Overclaiming, missing evidence, claims stronger than the data
- But what about **Authenticity**: is the evidence real?
  - Fabricated surveys, interviews, logs, benchmark runs
  - Needs things outside the paper: raw data, scripts, provenance, replication
  - But even with full replication package it may not be enough

JAMA Ophthalmology

Research Letter

FREE

Large Language Model Advanced Data Analysis Abuse to Create a Fake Data Set in Medical Research

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2023;141;(12):1174-1175. doi:10.1001/jamaophthalmol.2023.5162

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NEWS | 22 November 2023

ChatGPT generates fake data set to support scientific hypothesis

Researchers say that the model behind the chatbot fabricated a convincing bogus database, but a forensic examination shows it doesn't pass for authentic.

Even honest studies don’t reproduce

Reflections on the Reproducibility of Commercial LLM Performance in Empirical Software Engineering Studies

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Even honest studies don’t reproduce

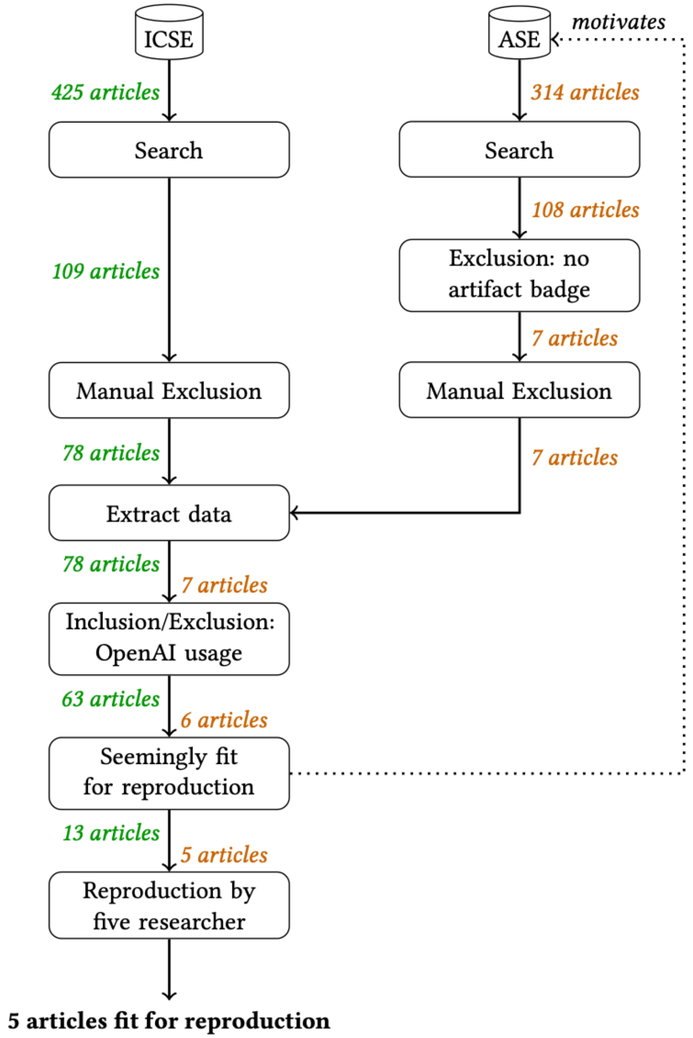


Figure 1: Summary of Research Approach

RQ1 Summary

Of 69 studies, only five were fit for reproduction. None of those five studies could be completely reproduced. Two studies could be partially reproduced, and three studies could not be reproduced.

RQ2 Summary

The main factors impeding the reproduction of LLM-centric empirical SE studies are missing or incomplete artefacts, missing details in reporting, dependency version issues, and, the only commercial LLM-specific factor: deprecated models.

RQ3 Summary

One year after the artefacts have been evaluated and awarded with the *Artifacts Evaluated - Reusable v1.1* badges, half of them do not fulfill (anymore) the requirements set up by ACM. This suggests that ACM artefact badges are not a reliable indicator of reproducibility.

Fake data, fragile results!? what should we require?

Let's discuss!

1. Split into groups that take on one of: surveys, interviews, repository or log data, benchmarks and tool evaluations. Each groups discusses and answers:

- If someone faked this kind of data, how would a reviewer tell?

- What one practice would make it more trustworthy, and what does it cost or fail to catch?

2. Each group reports back to full class

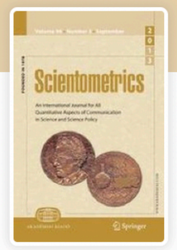
3. Discussion: Should the community require this?

Definitions and background

- *Scientometrics* is the science of measuring and analyzing science.
- *Bibliometrics* is the science of measuring and analyzing publications.
- Scientometrics is often done by applying bibliometrics on scientific publications.

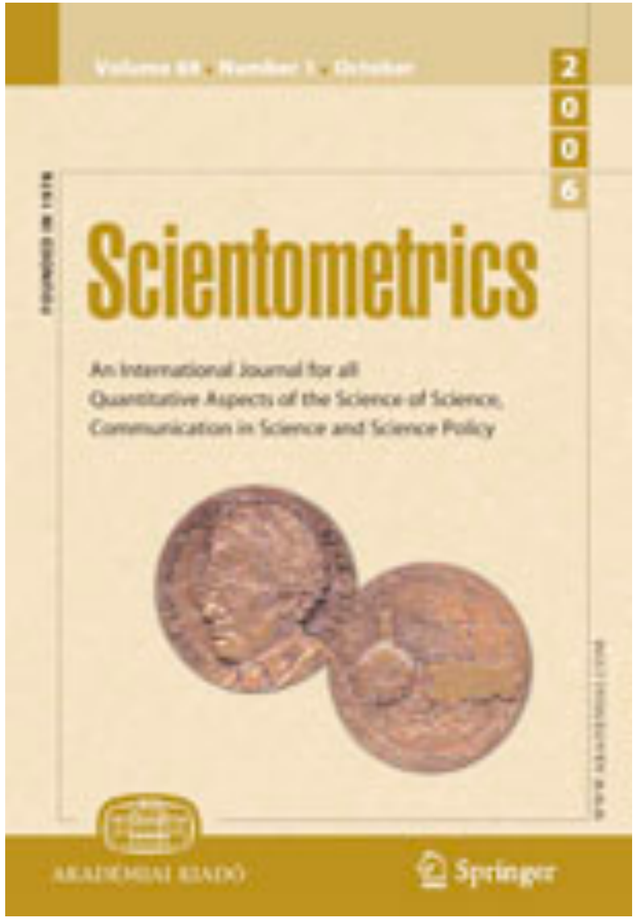
- There are of course scientific journals in this research area

Springer



Scientometrics

An International Journal for all Quantitative Aspects of the Science of Science, Communication in Science and Science Policy



Motivation (University perspective)

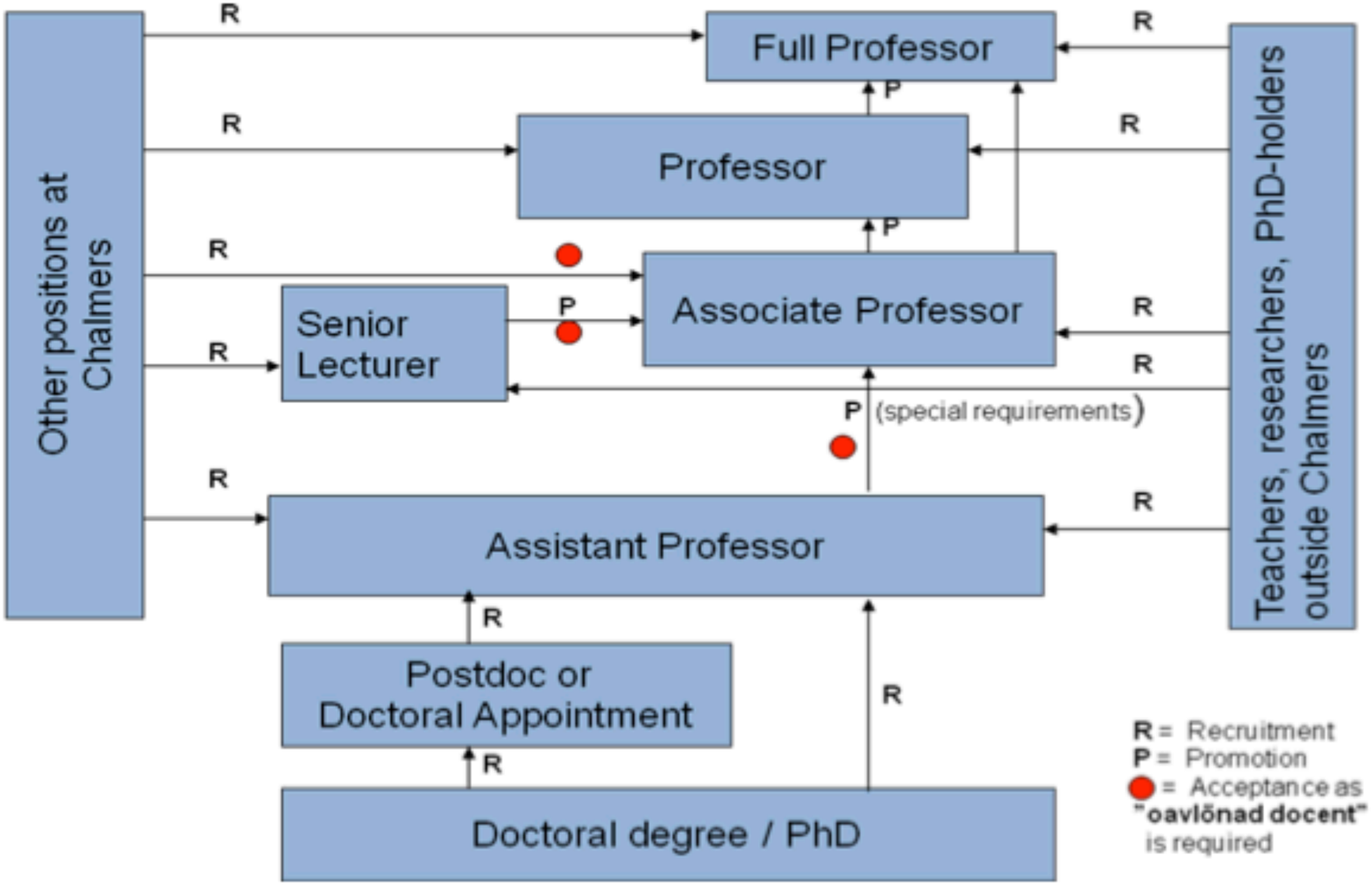
- Swedish Science Council (VR) wants everyone to state number of citations for every listed paper, but also says:

”Bibliometriska data, till exempel i form av det totala antalet citeringar för en sökande, bör användas med försiktighet i beredningsarbetet. Det har visat sig att sådana kan bli missvisande, om inte ett omfattande arbete med kvalitetssäkring utförts. Dock ska de sökande fr o m 2009 själva rapportera antalet citeringar för sina publikationer i ansökan.”

Motivation (individual perspective)

- Better understanding of ”system”
  - How are researchers being evaluated/ranked?
  - How are journals and conferences ranked?
- Can be useful e.g. when
  - selecting where to publish your research results
  - improving the impact of your research
  - applying for positions
  - applying for project funding
- However, trend seems to be somewhat away from explicit use of metrics
  - But implicit use still abundant or even seen as the normal state!?

Motivation (individual perspective)



University appointment regulations for teaching and research faculty at Chalmers

2011

Docent at Chalmers: What is assessed?

- **1. Scientific independence**: a documented ability to formulate and pursue your own research questions. You need **substantial personal contributions** and an **independent line of research beyond your PhD** and your former supervisors' work.
- **2. Scientific contributions**: research of sufficient **originality, significance, quality, breadth, depth and volume beyond the doctoral level**, judged in light of the field's conditions.
- **3. Teaching skill**: the ability to teach and supervise, especially at the advanced and doctoral levels, plus relevant training in university teaching.
- *Publication counts alone are not enough.*
  - The formal rules (handläggningsordningen) set only a minimum.
- Can strengthen the case: leadership, collaboration with society and industry, and other academic merits.
- Main evidence for **independence**: documented scientific work.

Metric for publications: Impact Factor et al

- The average number of cites to articles in a journal/proceedings in the last 2 years (-1)
- PageRank (similar to Google's algorithm) is a recursive impact factor, to give citations from journals that have high impact greater weight than citations from low-impact journals
- The number of citations to papers in a particular journal does not really directly measure the true quality of a journal, much less the scientific merit of the papers within it.
- Cannot be compared directly between different research areas/disciplines. It depends on, e.g.:
  - the absolute number of researchers in the area,
  - the average number of authors on each paper,
  - the nature of results in different research areas,
  - citation habits between different disciplines, particularly the number of citations per paper
- Review/survey papers/journals always get a lot of citations
- Venue h-index (like GScholar's area pages) depend heavily on size of venue
- Luckily, trend seems to be away from venue-specific rankings
  - But many countries still have requirements on publication in specific "quality tiers"

- Core general recommendation: "**do not use journal-based metrics such as the JIF as a surrogate** for the quality of individual articles, individual scientists' contributions, or hiring, promotion and funding decisions”
- 17 detailed recommendations addressed to 5 audiences, examples:
  - Funders and institutions:
    - be explicit about criteria; **content > metrics or venue**, especially for early-career researchers
    - value **all outputs, including datasets and software**
  - Publishers:
    - offer and promote article-level metrics
    - require author-contribution statements (CRediT)
  - Researchers:
    - judge by content when on committees;
    - use a range of article-level indicators
- Limits of DORA:
  - a declaration, with no enforcement or monitoring (CoARA tries to address this)

CoARA
Coalition for Advancing Research Assessment

- CoARA = Coalition for Advancing Research Assessment
- https://www.coara.org/
- Assessment should recognise **diverse** outputs, practices and activities.
- It should be based **primarily on qualitative judgement, with peer review central**, supported by responsible use of indicators.
- Timeline: an action plan within **1 year** of signing; full review cycle within **5 years.**

Universitetet i Bergen
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The University of Oslo
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Mid Sweden University
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Norwegian University of Science and Technology (NTNU)
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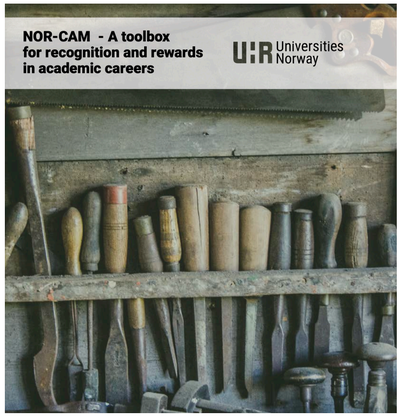
Luleå University of Technology
Website

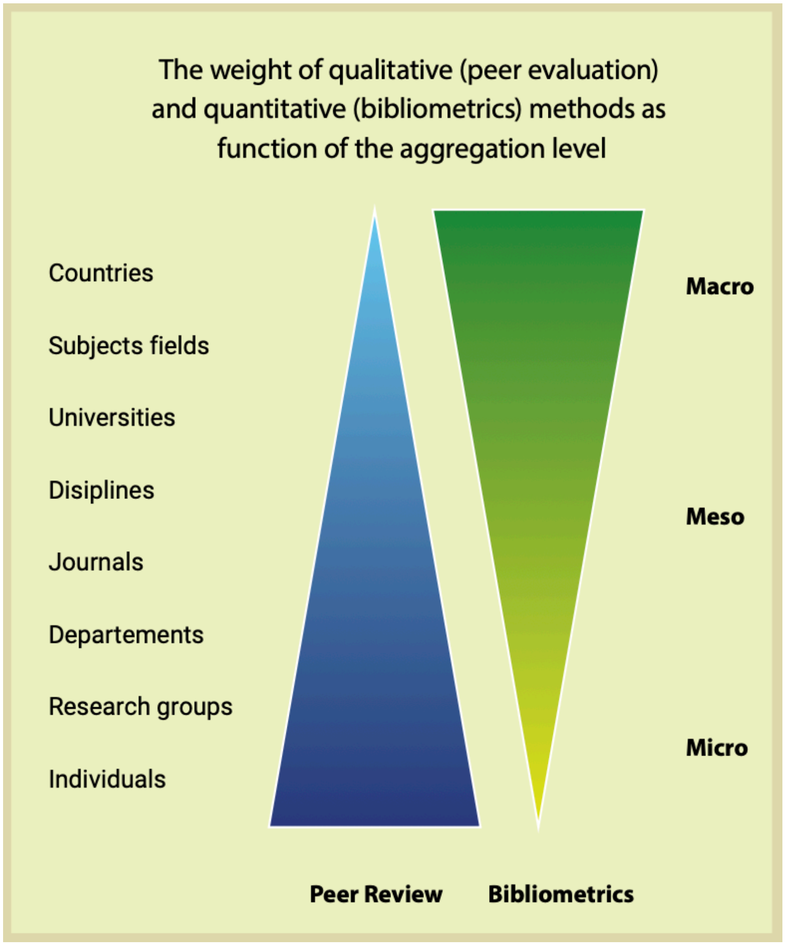
NOR-CAM - A toolbox for recognition and rewards in academic careers

UHR Universities Norway



- Bibliometrics must not be used in isolation
- Bibliometrics do not look ahead
- Bibliometrics do not understand varying contexts
- Bibliometrics cannot make decisions





Kanalregisteret

Academic publishing is no longer part of the funding of higher education institutions in Norway

From 2025 the Norwegian Government makes changes in the system for funding of Universities and University Colleges. Publication points will no longer be part of the funding of the higher education institutions in Norway.

Published: 18. September 2024

Meanwhile in Sweden… ;)

Vetenskapsrådet

På svenska

Start / Just now / News

NEWS

Publications and external funding to play a greater role in direct government grants

The Government is changing its model for allocating direct grants to higher education institutions (HEIs). The impact of the research and the HEIs' ability to attract external funding will be rewarded more than previously. The Swedish Research Council has been tasked by the Government to develop a proposal ahead of the change, which will come into force in 2026.

AI and your research career

Let's discuss!

1. You are on a hiring committee in 2030. Everyone has 3x more papers than today. What do you look at when assessing candidates?

3 recommendations:

R1. Develop and evolve your **AI-use policy**! Template:
- What I use AI for (research, writing, reviewing)
- What I never delegate, and why
- How I check AI output
- What I disclose, where and to whom (paper, co-authors, CRediT, venue policy)
- Confidentiality: what I never upload (e.g. papers I review, others' unpublished work, participant data)

R2. Design and write down your **Publication strategy**!

R3. Design and write down your **Research Method strategy**!

Feedback, criticism/praise, ideas for improvement,
Please email:

robert.feldt@chalmers.se

Thank you!